Neural Networks

DSN: Energy-Efficient EMG Signal Classification Method Enabling Real-Time Monitoring for Edge Healthcare

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Abstract:	Surface electromyography (sEMG) signal classification is challenged by high computational cost and energy consumption, which limits its deployment on resource-constrained wearable devices. To address these issues, we propose DSN, a hybrid framework that combines Spiking Neural Networks (SNNs) with Hyper-dimensional Computing (HDC). The SNN module adopts randomized initialized weights with lightweight one-shot forward-backward training for efficient feature extraction, while the HDC module leverages high-dimensional distributed representations for robust classification. Based on theoretical derivation, we've proved its noise robustness, efficiency and hardware error robustness. Experimental results show that our method achieves up to 12.03× lower testing energy consumption compared with LSTM and 3.46× lower compared with GRU, while maintaining over 90% accuracy using only 20% of the training data. Moreover, the framework exhibits strong robustness to noise and hardware faults, making it highly suitable for real-time, energy-efficient sEMG monitoring on edge healthcare platforms.